

Deep Learning (285652)

ClaimGuard: An Evidence-Backed Pipeline for Academic Citation Integrity Checking

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1. Abstract

Academic integrity checking requires more than detecting copied text: a reference may be genuine while failing to support the sentence that cites it. ClaimGuard identifies claim-bearing sentences and missing citations, validates references, retrieves evidence, and assigns five support labels. Its review interface preserves the evidence trail and compares verifier backends on the same input. The implementation combines transparent rules, scholarly metadata APIs, lexical or Sentence-BERT retrieval, local Ollama, an OpenAI frontier model, and a SciFact LoRA classifier. On a balanced 30-case support benchmark, lexical verification achieved 0.900 accuracy and Macro-F1; embedding retrieval reached 0.867 accuracy and 0.868 Macro-F1. On 50 manually annotated sentences from five reports, claim-type accuracy fell to 0.520 and citation-needed F1 to 0.267, revealing domain shift. A DistilBERT LoRA adapter achieved 0.353 Macro-F1 on SciFact validation but only 0.172 after transfer to ClaimGuard’s incompatible five-label taxonomy. A one-claim frontier/local diagnostic also produced different verdicts. ClaimGuard is therefore best used for auditable reviewer triage, not automatic misconduct decisions. Principal limitations are noisy PDF extraction, small single-annotator datasets, incomplete source access, API failures, model non-determinism, and limited hardware.

Keywords: academic integrity, citation verification, retrieval-augmented generation, scientific claim verification, LoRA, human-in-the-loop review

2. Introduction

Academic reports increasingly combine conventional references, web-accessible research, and language-model-assisted writing. A bibliography may contain malformed or fabricated records, but even a genuine paper may not establish the claim attached to it. Plagiarism detection addresses textual overlap; it does not determine whether a sentence needs a citation, whether the cited work exists, or whether available evidence supports the sentence. Manually answering these questions requires repeated movement between a report, bibliographic databases, and source texts, which encourages superficial spot checks as documents grow.

The risk is asymmetric. Missing a misleading citation weakens assessment quality, while a false positive may unfairly imply misconduct. ClaimGuard consequently frames its output as decision support. It prioritizes passages for inspection and exposes the sentence, citation marker, parsed metadata, retrieved evidence, verifier, confidence, latency, and rationale. No score is converted into an accusation.

The project investigates four questions: (RQ1) Can lightweight section-aware rules identify claim types and missing citations in authentic PDFs? (RQ2) Can IEEE-like references be recovered and validated despite PDF extraction errors? (RQ3) How well do lexical and semantic retrieval support a five-way source-support judgment? (RQ4) What trade-offs emerge among a transparent heuristic, local Ollama, an OpenAI frontier model, and a SciFact-trained LoRA

adapter?

All three core modules were implemented: claim and citation analysis, reference parsing and validation, and retrieval-based support verification. Optional work comprises interchangeable verifier backends, Sentence-Transformer retrieval, a GPU-trained LoRA adapter, a Streamlit comparison interface, and an AI-text detection interface. The calibrated Binoculars model pair was not run because two Falcon-7B models exceed the available 8 GB GPU. The contribution is thus not a single classifier, but an inspectable connection between sentence-level citation need, bibliographic identity, evidence, and alternative verifier backends.

3. Related Work

SciFact introduced expert-written scientific claims with support, refutation, and evidence rationales [1]. MultiVerS extended scientific verification with weak supervision and full-document context [2]. Both show why absence of evidence must remain distinct from contradiction. ClaimGuard adds upstream citation-need detection and bibliographic identity checks to this verification setting.

Citation-needed detection is a distinct upstream task. Redi et al. developed a citation-need taxonomy and models for Wikipedia verifiability [12]; their domain and editorial conventions differ from student reports. Sarol et al. paired retrieval with verification to study subtle citation-integrity errors in biomedical publications [13]. ClaimGuard targets the same claim–evidence relationship but adds local document ingestion and metadata validation.

Retrieval-augmented generation links models to external evidence [3], while Sentence-BERT supports efficient semantic retrieval [4]. GROBID and S2ORC provide document and bibliographic infrastructure [5, 6]; OpenAlex and Crossref expose complementary scholarly metadata [7, 8]. scite classifies citation contexts in indexed literature [9], whereas ClaimGuard starts from an arbitrary local report and connects its sentence to evidence.

LoRA enables parameter-efficient adaptation [10]. Binoculars provides a model-pair-dependent AI-text signal [11], but paraphrasing and distribution shift undermine detector reliability [14]; broad tool tests likewise caution against using such scores as misconduct evidence [15]. ClaimGuard therefore treats AI detection as optional risk signaling. Its gap is integration: citation need, bibliographic identity, retrieval, verifier disagreement, and a reviewer-facing evidence trail in one workflow.

4. System Architecture and Methods

4.1. Pipeline

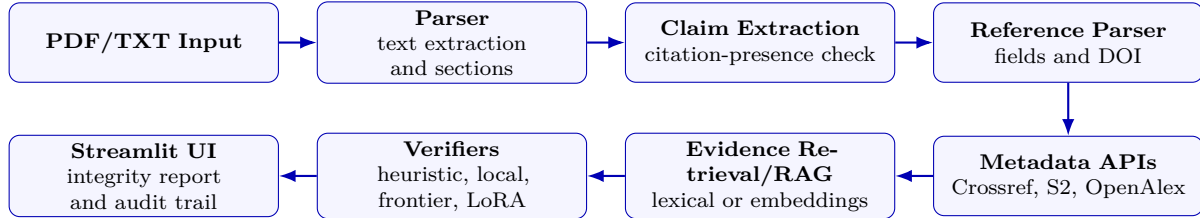


Figure 1: ClaimGuard architecture and data flow. Evidence, confidence, latency, and backend metadata remain attached to the final report.

Core module	Implemented components	Output	Evaluation evidence	Limitations
1: Claim extraction and citation presence	Section-aware sentence types, citation linking, inherited citations, severity rules	Sentence label, cited indices, missing-citation flag/severity	50 real sentences: accuracy .520, Macro-F1 .437; citation-needed F1 .267	Single annotator; rules confuse definitions, methods, and interpretations
2: Bibliographic validation	IEEE-like parser; title/author/year/venue/DOI matching; Crossref, Semantic Scholar, OpenAlex; partial retraction flags	Parsed record, candidate, DOI, confidence, validation/problem status	Five-record parser regression: all fields 1.000; metadata baseline: 1/5 exact top-1	No gold evaluation of DOI resolution or retraction detection; API/access failures
3: Deep claim-source verification/RAG	Chunking, lexical/embedding retrieval, heuristic/Ollama/OpenAI/LoRA interfaces	Evidence chunks, five-way verdict, rationale, backend metadata	30 synthetic cases: lexical Macro-F1 .900; embeddings .868	Synthetic short passages; incomplete full text; only one frontier/local diagnostic

Table 1: Core Module Compliance. Metrics refer only to the listed saved evaluations.

The parser accepts PDF and plain text, cleans extraction artifacts, separates body, references, and supplied evidence, and propagates section headings. Sentence splitting protects initials, abbreviations, decimal values, and IEEE citation markers. The claim module assigns one of five types: factual claim, methodological statement, opinion/interpretation, background/definition, or non-claim. Rules combine empirical verbs, numerical patterns, method cues, hedging, and definitional constructions. A factual sentence outside methods or results is flagged when it lacks a same-sentence or safely inherited citation; severity rises for numerical, causal, universal, or explicit prior-work assertions.

The reference module extracts index, authors, title, venue, year, and DOI and repairs common PDF artifacts. A parsed DOI is normalized and used for DOI-specific Crossref, Semantic Scholar, and OpenAlex lookups before title search; matching scores combine DOI, title, author, year, and venue. Crossref retraction relations/subtypes and OpenAlex’s `is_retracted` field can produce `retracted_or_problematic`. This is partial coverage: Semantic Scholar candidates do not

contribute a retraction flag, DOI resolution is database-based rather than a complete DOI-registry audit, and neither DOI validity nor retraction detection has a gold-labeled evaluation. Candidate metadata remains visible rather than being silently accepted.

Evidence from abstracts, supplied passages, metadata, or accessible full text is split into overlapping chunks. Lexical retrieval normalizes tokens, removes stop words, lightly stems terms, and scores overlap. Semantic retrieval encodes claims and chunks with `all-MiniLM-L6-v2` and normalized cosine similarity. Retrieval is restricted to the cited reference when possible, reducing contamination from unrelated works.

4.2. Verifier backends and interface

The deterministic verifier uses coverage, negation, directional conflict, and explicit absence-of-evidence cues. Ollama sends the same claim and top evidence to `qwen3:1.7b` at temperature zero. The OpenAI backend uses `gpt-5.4-mini`, disables response storage, and requires structured JSON. Both return exactly one of *supported*, *partially supported*, *not supported*, *contradicted*, or *insufficient evidence*, plus confidence and rationale. The shared prompt instructs models to judge only the supplied evidence; its full text and a sanitized response are in Appendix A.

The local SciFact classifier starts from `distilbert-base-uncased` with rank-8 LoRA adapters. It was trained on 1,261 pairs and validated on 450 pairs. Its native three labels map to only three of ClaimGuard’s five categories, making transfer evaluation intentionally difficult. The Streamlit interface presents document priority, sentence-level findings, evidence, backend identity, rationale, latency, and errors. Its comparison view aligns predictions for identical inputs. Agreement is not called accuracy when no gold label exists, and confidence values from different backends are not treated as calibrated equivalents.

4.3. Evaluation design

Evaluation is separated into authentic-document evidence and controlled regression evidence. The authentic set contains ten manually selected sentences from each of five course reports ($N = 50$), labeled once for claim type and citation need; no inter-annotator agreement is claimed. Five Report-5 references test parsing and a manual Crossref top-1 baseline. Controlled tests contain 25 synthetic Module-1 patterns and 30 balanced synthetic claim–evidence pairs (six per support label). SciFact validation uses 450 native pairs. The OpenAI/Ollama result is a one-claim execution diagnostic without a gold label, not an accuracy benchmark. This separation prevents synthetic scores from being presented as real-document performance.

5. Results

5.1. Quantitative results

Component	N	Acc.	Prec.	Recall	F1
<i>Synthetic/controlled regression evidence</i>					
Synthetic citation-needed detection	25	1.000	1.000	1.000	1.000
Five-way verification, lexical	30	0.900	0.905	0.900	0.900 ^a
Five-way verification, embeddings	30	0.867	0.900	0.867	0.868 ^a
SciFact LoRA validation	450	0.409	–	–	0.353 ^a
LoRA transfer to five labels	30	0.300	0.130	0.300	0.172 ^a
<i>Authentic-document evidence</i>					
Real claim-type classification	50	0.520	–	–	0.437 ^a
Real factual-claim detection	50	–	0.455	0.833	0.588
Real citation-needed detection	50	–	0.250	0.286	0.267
IEEE reference-field parsing	5	1.000	–	–	1.000 ^b

^aMacro-F1; ^bcomplete-record accuracy. No P/R was produced for exact field matching.

Table 2: Saved evaluation results, explicitly separated by evidence type.

The dominant finding is the synthetic-to-real gap. Handcrafted patterns were perfect on the 25-case regression set, yet real-report claim accuracy was 0.520. Definitions were never recovered correctly as a class, and nine of fifteen methodological statements became factual claims. Citation-needed recall was 0.286: five of seven annotated omissions were missed. Rules remain useful for prioritization but do not generalize sufficiently for automatic enforcement.

Gold \ predicted	Factual	Method	Opinion	Definition	Non-claim
Factual claim	15	0	2	0	1
Methodological statement	9	6	0	0	0
Opinion/interpretation	4	0	5	0	1
Background/definition	5	0	0	0	2

Table 3: Confusion matrix for the 50 authentic sentences. No gold examples were labeled non-claim.

Module-2 check	Implementation	Saved evaluation evidence	Reportable conclusion
Field and DOI extraction	Regex parsing and normalization	$N = 5$: DOI-field and all-field accuracy 1.000	Small parser regression only
DOI validity/resolution	DOI-specific queries to three meta-data APIs	No gold-labeled DOI-resolution output	Implemented, not evaluated
Retraction status	Crossref relation/subtype and OpenAlex flag	No gold-labeled retraction output	Partial implementation, not evaluated

Table 4: Module-2 evidence boundary. Missing metrics are not inferred from parser accuracy.

On the balanced support benchmark, lexical verification reached 0.900 Macro-F1 and perfect contradiction F1. Embeddings reduced Macro-F1 to 0.868 because the short benchmark passages already contain highly diagnostic wording. The LoRA model reached 0.353 Macro-F1 on native SciFact validation and 0.172 after five-label transfer. It cannot emit *partially supported* or *not supported*, which together form 12 of 30 transfer cases.

5.2. Qualitative behavior

Case	Observed behavior	Interpretation
RQ5 sentence 12	Flagged an uncited claim that retrieval is a central RAG bottleneck.	Useful high-severity triage.
Report-5 references	Recovered five titles, years, and venues despite collapsed spacing.	Parser repair succeeded on authentic PDF text.
Contradiction C1	Matched “do not eliminate errors” against “eliminate all hallucinations.”	Negation and direction cues worked.
RQ1-02	Labeled a result interpretation as a factual claim.	Mixed rhetorical roles confuse rules.
RQ5 title page	Flagged header text as an uncited factual claim.	Layout noise survived extraction.

Table 5: Five representative successes and failures.

5.3. Local versus frontier diagnostic

Backend	N	Gold Acc./F1	Verdict / agreement	Latency	Tokens / cost	Observed behavior
OpenAI gpt-5.4-mini	1	– / –	insufficient evidence / disagree	1.889 s	441 input + 65 output; monetary cost not logged	Withheld support because evidence omitted the “without changing weights” clause
Ollama qwen3:1.7b	1	– / –	supported / disagree	40.902 s	388 prompt + 447 generated; no API cost logged	Accepted support although the retrieved metadata did not state the full clause

Table 6: One-claim execution diagnostic on identical retrieved evidence. There is no gold label, accuracy estimate, or general cost comparison.

Only one pair contains actual model calls. A second aligned output was labeled *insufficient evidence* before invocation (`model=null`) and is excluded. Thus the frontier model was faster and more conservative in this case, but no superiority or typical failure mode can be inferred. A full gold-labeled comparison remains future work.

5.4. Metadata-search baseline

#	Queried Report-5 work (short title)	Score	Crossref top-1 judgment	judgment
1	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	.593	Incorrect variant	later QA
2	Sentence-BERT	.867	Correct	
3	RAG for Large Language Models: A Survey	.780	Incorrect variant	graph-RAG
4	Lost in the Middle	.779	Incorrect	“Found in the Middle”
5	LLMs Distracted by Irrelevant Context	.459	Incorrect	unrelated security work

Table 7: Five-case Crossref metadata-search baseline. Scores are fuzzy title similarity.

Crossref’s top candidate was exact for one of five records. Two incorrect near-neighbors exceeded the nominal 0.75 fuzzy threshold because title terms and year overlapped. This metadata-search baseline does not evaluate citation need or claim–source support. No completed scite, SemanticCite, or other external-tool prediction file exists in the repository; the available CSV is only a template. A full existing-tool benchmark therefore remains future work. ClaimGuard should be understood as an orchestration layer over scholarly services, not their replacement.

5.5. Answers to the research questions

RQ	Evidence-grounded answer
RQ1	Only partially. On 50 authentic sentences, claim-type accuracy was .520 (Macro-F1 .437) and citation-needed F1 was .267; lightweight rules are triage, not comprehensive detection.
RQ2	Parsing handled the five-record regression set (all-field accuracy 1.000), but Crossref top-1 identity was exact for only 1/5 real references. DOI resolution and retraction detection lack gold evaluation.
RQ3	On 30 controlled cases, lexical verification reached .900 Macro-F1 and embeddings .868. Because evidence was short and synthetic, this does not establish real-document verification accuracy.
RQ4	The heuristic led the controlled five-way test; LoRA transfer reached .172 Macro-F1. One no-gold OpenAI/Ollama disagreement is insufficient to rank local and frontier models or justify general cost claims.

Table 8: Direct answers to RQ1–RQ4.

6. Discussion

ClaimGuard’s strongest property is its end-to-end audit trail: a reviewer can move from sentence and citation marker to metadata candidate, retrieved passage, backend, and rationale. The main empirical lesson is nevertheless caution. Strong controlled verification (.900 Macro-F1) coexists with weak real-document claim transfer (.437 Macro-F1), so synthetic regression scores must not be read as deployment performance.

Failure mode	Cause	Example/evidence	Mitigation
Layout false positive	PDF headers and title text enter sentence stream	RQ5 title/header flagged as an uncited claim	Layout-aware parsing; exclude front matter
Claim-type confusion	Mixed rhetorical roles and narrow rules	9/15 method statements predicted factual	Learned section-aware model with abstention
Metadata near-neighbor	High title/year overlap	“Lost in the Middle” matched to “Found in the Middle”	Require DOI/author agreement; show candidates
Access bias	Paywall/non-OA or abstract-only evidence	No source-level rate measured; missing full text can yield insufficient evidence	Separate access failure from support; licensed/OA retrieval
Taxonomy mismatch	SciFact has three labels, ClaimGuard five	LoRA transfer Macro-F1 .172	Align labels before fine-tuning

Table 9: Observed or explicitly unmeasured failure modes and mitigations.

Access bias deserves special emphasis: paywalled or non-open-access works are harder to verify than sources with complete machine-readable text. Consequently, *insufficient evidence* may describe retrieval or access limits rather than an unsupported claim, and may systematically favor open-access literature. API timeouts and incomplete deposits add similar missingness. The evaluation itself is small and single-annotator; DOI/retraction behavior and full local/frontier accuracy remain unevaluated.

False positives are consequential in academic assessment. Common knowledge, inherited citations, extraction errors, and cautious scientific language can all trigger incorrect flags. AI-text scores are particularly unsuitable as authorship proof [14, 15]. ClaimGuard therefore avoids misconduct labels and requires inspection of the original source. Future work should add double annotation, calibrated abstention, evidence-span labels, stricter metadata identity tests, gold DOI/retraction cases, reviewer studies, a completed external-tool benchmark, and a 30-case local/frontier evaluation with recorded monetary cost. Until then, ClaimGuard is transparent triage, not automatic enforcement.

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A. Reproducibility and Prompt Details

A.1. Environment and saved outputs

Experiments used Python 3.14.3, PyTorch 2.11.0 with CUDA 12.8, an NVIDIA RTX 3070 Ti with 8 GB VRAM, and seed 42 for LoRA training. Primary artifacts are:

- `rq_claim_evaluation.json`
- `reference_evaluation.json`
- `evaluation_lexical.json`
- `evaluation_embeddings.json`
- `evaluation_lora.json`
- `model_comparison.json`
- `crossref_baseline_rq5.json`
- `scifact-lora/training_report.json`

API identifiers are retained locally; credentials are excluded.

A.2. Verifier prompt

Classify the claim using exactly one status: `supported`, `partially_supported`, `not_supported`, `contradicted`, or `insufficient_evidence`. Judge only from the supplied evidence. Return JSON with status, confidence (0..1), and a concise rationale.

A.3. LoRA and execution details

The adapter used rank 8, alpha 16, query/value projections, an effective batch size of 16, three epochs, and 740,355 trainable parameters (1.09% of the model). Reproduction commands and additional API response samples are stored with the project outputs rather than repeated in the main report.